

Investigation 2

Context: Your team has informed you that the adversary has managed to set up persistence on your endpoints as they continue to move throughout your network. Find how the adversary managed to gain persistence using logs provided.

File path: C:\Users\user\Documents\TryHackMe!\Sysmon\investigation 2.

I'll use .envtx file to find clues and respond to the answers.

Question1.

What is the full path of the payload in Investigation 2?

Here I'm looking for executed command or process, I started by reviewing the file

```
PS C:\Users\user> Get-WinEvent -Path '.\Documents\TryHackMe!\Sysmon\investigation 2\Investigation-2.envtx'
```

ProviderName: Microsoft-Windows-Sysmon

TimeCreated	Id	LevelDisplayName	Message
6/15/2019 1:14:32 AM	1		
6/15/2019 1:13:44 AM	3		
6/15/2019 1:13:42 AM	1		

The command shows two 1 Symon IDs that indicate process creation, I began looking there!

```
PS C:\Users\user> Get-WinEvent -Path '.\Documents\TryHackMe!\Sysmon\investigation 2\Investigation-2.envtx' `
>> -FilterXPath "[*][System[(EventID=1)]] and *[EventData[Data[@Name='Image']='C:\Windows\System32\mshta.exe']]" |
>> ForEach-Object {
>>     [xml]$xml = $_.ToXml()
>>     $xml.Event.EventData.Data | Where-Object {$_.Name -eq "CommandLine"} | Select-Object -ExpandProperty '#text'
>> }
"C:\Windows\System32\mshta.exe" "C:\Users\IEUser\AppData\Local\Microsoft\Windows\Temporary Internet Files\Content.IE5\S97WTYG7\update.hta"
```

Filtering by ID "1" I get the process suspicious "mshta.exe" this process is often used by the attackers to run malicious HTML applications, and in the marked section underline is the path of the payload used by the attacker.

Question 2.

What signed binary executed the payload in Investigation 2?

I know that ID 1 is a process creation, so I start for there, the process is running some file .hta according to above snippet.

Now we filter these two clues

```
PS C:\Users\User> Get-WinEvent -Path '.\Documents\TryHackMe!\Sysmon\investigation 2\Investigation-2.evtx' `
>> -FilterXPath "[*][System/EventID=1]" |
>> ForEach-Object {
>>     [xml]$xml = $_.ToXml()
>>
>>     $cmd = $xml.Event.EventData.Data | Where-Object { $_.Name -eq "CommandLine" } | Select-Object -ExpandProperty '#text'
>>
>>     # Look for update.html or update.hta anywhere in the string
>>     if ($cmd -match "update\.(html|hta)") {
>>         $image = $xml.Event.EventData.Data | Where-Object { $_.Name -eq "Image" } | Select-Object -ExpandProperty '#text'
>>     }
>> }
>> }
C:\Windows\System32\mshta.exe
```

Now I extract the binary is “mshta.exe”.

Let’s move to the next one

Question 3.

What is the IP of the adversary in Investigation 2?

This is easily question, I used connection id = 3 and filtered by Destination IP matches with the malicious binary “mshta.exe”

```
PS C:\Users\User> Get-WinEvent -Path '.\Documents\TryHackMe!\Sysmon\investigation 2\Investigation-2.evtx' `
>> -FilterXPath "[*][System/EventID=3]" |
>> ForEach-Object {
>>     [xml]$xml = $_.ToXml()
>>
>>     $image = $xml.Event.EventData.Data | Where-Object { $_.Name -eq "Image" } | Select-Object -ExpandProperty '#text'
>>     $dstIp = $xml.Event.EventData.Data | Where-Object { $_.Name -eq "DestinationIp" } | Select-Object -ExpandProperty '#text'
>>
>>     if ($image -match "mshta.exe") {
>>         $dstIp
>>     }
>> }
>> }
10.0.2.18
PS C:\Users\User>
```

That’s how I got the ip address of the malicious actor.

Next question.

What back connect port is used in Investigation 2?

This is very similar to the last one, I used the same ID = 3 to looking for the port connection related with the binary “mshta.exe”

```
PS C:\Users\user> Get-WinEvent -Path '.\Documents\TryHackMe!\Sysmon\investigation 2\Investigation-2.evtx' `
>> -FilterXPath "[*System/EventID=3]" |
>> ForEach-Object {
>>     [xml]$xml = $_.ToXml()
>>
>>     $image = $xml.Event.EventData.Data | Where-Object { $_.Name -eq "Image" } | Select-Object -ExpandProperty '#text'
>>     $dstip = $xml.Event.EventData.Data | Where-Object { $_.Name -eq "DestinationIp" } | Select-Object -ExpandProperty '#text'
>>
>>     if ($image -match "mshta.exe") {
>>         $dstip
>>     }
>> }
10.0.2.18
PS C:\Users\user>
```

And I got it!... so, I finish this investigation and move to next one.